Hypothesis Testing.

Many Engineering problems require that we decide whether to accept or reject a statement about some parameter. The statement is called a hypothesis & the decision making procedure about the hypothesis is called hypothesis testing.

A statistical hypothesis is a statement about the parameters of one or more populations.

Since we use prob. distn to represent populations, a statistical hypothesis may also be thought of as a statement about the prob. distn. of a random variable. The hypothesis will usually involve one or more parameters of this distribution.

For eg, we are interested in burning rate of a solid propellant used to power aircrew escape systems. Now burning rate is a random variable than can be described by a probability distribution. If our interest focuses on mean burning rate (a parameter of this distn). To be specific, we are interested in deciding whether or not the mean burning rate is 50 cm/sec. We may express this as

$H_0: \mu = 50$ cm/sec.
$H_1: \mu \neq 50$ cm/sec.

The statement $H_0: \mu = 50$cm/sec is called null hypothesis. and the statement $H_1: \mu \neq 50$cm/sec is called alternative hypothesis. Since the alternate hypothesis specifies values of μ that could be either $>$ or $<$ than 50 cm/sec, it is called a two-sided alternative hypothesis. In some situations, we may wish to formulate a one sided alternative hypothesis as in

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H_0: $\mu = 50$ cm/sec or H_0: $\mu = 50$ cm/sec.
H_1: $\mu < 50$ cm/sec H_1: $\mu > 50$ cm/sec

* Hypothesis are always statements about the popalation or distribution under study, not statements about the sample.

The value of popln parameter specified in null hypothesis i.e (50 cm/sec) is usually determined in one of 3 ways:

1. Result from past experience or knowledge of previous tests or expts. then hypothesis objective and testing is done usually to determine whether the parameter value has changed.
2. This value may be determined from some theory or model regarding the process under study. Here the objective of hypothesis testing is to verify the theory or model.
3. Third situation is when the value of popln parameter results from external considerations, such as design or engg. specifications, or from contractual obligations. In this case, the objective of hypothesis testing is conformance testing.

A procedure leading to a decision about a particular hypothesis is called a test of a hypothesis. Hypothesis-testing procedures rely on using the infn in a random sample from the population of interest. If this infn. is consistent with the hypothesis, we will not reject the hypothesis; however, if this information is inconsistent with the hypothesis, we will conclude that the hypothesis is false.

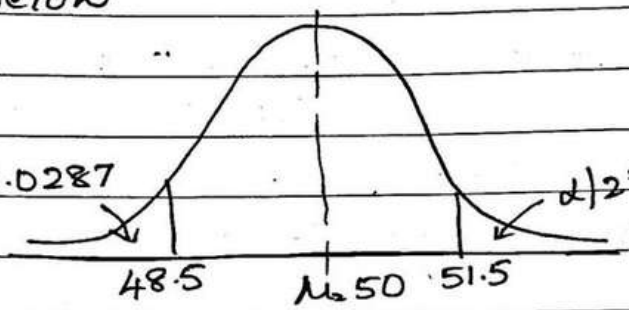
In the previous example, suppose that a sample of $n = 10$ specimens is tested & sample mean' burning rate $\bar{x}$ is observed. The sample mean is an estimate of true popula-tion mean μ. A value of the sample mean $\bar{x}$ that falls close to the hypothesized value of $\mu = 50$cm/sec does not conflict with null hypothesis that the true mean μ is really 50cm per sec. On the other hand, a sample mean that is considerably diff. from 50 cm/sec is evidence in support of

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the alternate hypothesis H_1. Thus the sample mean is the test statistic in this case.

The sample mean can take on many diff. values. Suppose that if $48.5 \leq \bar{x} \leq 51.5$, we will not reject the null hypothesis $H_0: \mu = 50$ & if either $\bar{x} < 48.5$ or $\bar{x} > 51.5$, we will reject the null hypothesis in favor of alternate hypothesis $H_1: \mu \neq 50$ as illustrated below



Critical region for $H_0: \mu = 50$ versus $H_1: \mu \neq 50$ & n=10

The values of $\bar{x}$ that are < 48.5 & > 51.5 constitute the critical region for the test, while all values that are in the interval $48.5 \leq \bar{x} \leq 51.5$ form a region for which we fail to reject the null hypothesis. By convention, this is usually called the acceptance region. The boundaries b/w the critical regions & acceptance region are called the critical values. In this eg. 48.5 and 51.5 are critical values. It is customary to state conclusions relative to null hypothesis H_0. Therefore, we reject H_0 in favor of H_1 if the test statistic falls in the critical region & fail to reject H_0 otherwise.

This decision procedure can lead to either of 2 wrong conclusions.

Type I error :- Rejecting the null hypothesis H_0 when it is true is defined as Type I error.

Type II error :- Failing to reject the null hypothesis when it is false is defined as Type II error.

Decisions in Hypothesis testing.

Decision	H_0 is True	H_0 is False
Fail to reject H_0	no error	type II error
Reject H_0	type I error	no error

The probability of making a type I error is denoted by α.

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

Sometimes the type I error probability is called significance level, or the α-error or the size of the test

Note.
1. We can reduce α by widening the acceptance region.
2. We can also reduce α by increasing the sample size

The prob. of a type-II error is denoted by β.

i.e $\beta = P(\text{Type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$.

To calculate β we should have a specific alternative hypothesis i.e a particular value of μ for H_1.

Note: 1. The prob. of making a type II error β increases rapidly as the true value of μ approaches the hypothesized value.
2. The type II error prob. also depends on sample size n. Increasing the sample size results in decrease in prob of Type II error.

4 important points to ponder.

1. The size of the critical region and consequently the prob. of a type I error α, can always be reduced by appropriate selection of critical values.
2. Type I & type II errors are related. A decrease in prob of one type of error always results in an increase in the prob of

the other, provided that the sample size n does not change.

3. An increase in sample size reduces β, provided that α is held constant.

4. When the null hypothesis is false, β increases as the true value of the parameter approaches the value hypothesized in the null hypothesis. The value of β decreases as the diff b/w the true mean & hypothesized value increases.

Generally the analyst controls the type I error prob α when he or she selects the critical values. Thus it is easy for analyst to set Type I error prob at any desired value. Since analyst can directly control the prob. of wrongly rejecting H_0, we always think of rejection of null hypothesis H_0 as strong conclusion.

On the other hand the prob of Type II error β is not constant but depends on true value of the parameter. It also depends on sample size that we have selected. Since β depends or is fn. of both sample size & extent to which the null hypothesis H_0 is false, it is customary to think of decision to accept H_0 as a weak conclusion, unless we know that β is acceptably small. ∴ rather than saying "Accept H_0" we prefer the terminology "fail to reject H_0". Failing to reject H_0 implies that we have not found sufficient evidence to reject H_0, i.e to make a strong statement.

Failing to reject H_0 does not necessarily mean that there is a high probability that H_0 is true. It may mean that more data are required to reach a strong conclusion.

Power — The power of a statistical test is the prob of rejecting the null hypothesis, H_0, when alternate hypothesis is true.

The power is computed as $1-\beta$. & power can interpreted as the prob. of correctly rejecting a false null.

We often compare statistical tests by comparing their power differences.

One Sided & Two Sided Hypotheses

In formulating one sided alternate hypotheses, we should remember that rejecting H_0 is always a strong conclusion. Consequently we should put the statement about which it is important to make a strong conclusion in the alternate hypothesis. In real-world problems, this will often depend on our point of view & experience with the situation.

Pvalues in Hypothesis Tests.

The p-value is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

It is not always easy to compute the exact P-value for a test. However most modern computer programs for statistical analysis report P-values.

General Procedure for Hypothesis Tests.

1. From the problem context, identify the parameter of interest.
2. State the null hypothesis H_0.
3. Specify an appropriate alternative hypothesis H_1.
4. Chose a significance level α.
5. Determine an appropriate test statistic.
6. State the rejection region for the statistic. This step is not necessary if the p-value approach is used.
7. Compute any necessary sample quantities, substitute these into the equation for the test statistic, & compute that value.
8. Decide whether or not H_0 should be rejected and report that in problem context.

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Problems.

1. In each of the foll. situations, state whether it is a correct stated hypothesis testing problem & why.

(a) $H_0: \mu = 25$ $H_1: \mu \neq 25$ Yes. We will always state H_0 an equation

(b) $H_0: \sigma \geq 10$ $H_1: \sigma = 10$ No. Null Hypothesis - sp. value.

(c) $H_0: \bar{x} = 50$ $H_1: \bar{x} \neq 50$ No Hypothesis statements are for population & not for samples.

d $H_0: p = 0.1$; $H_1: p = 0.5$
No. values in both hypothesis are different.

e. $H_0: s = 30$ $H_1: s > 30$
Hypotheses are always statements about the popln or distn under study, not statement about the sample.

2. A semiconductor manufacturer collects data from a new tool & conducts a hypothesis test with the null hypothesis that a critical dimn mean width equals 100 nm. The conclusion is to reject the null hypothesis. Does this result provide strong evidence that the critical dimn mean equals 100 nm? Explain.

The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire population. This is usually impossible in most practical situations. Therefore, a hypothesis testing procedure should be developed with the prob. of reaching a wrong conclusion in mind.

3. The mean pull-off force of a connector depends on cure time.

a) State the null & alternate hypotheses used to demonstrate that the pull-off force is below 25 newtons.

b) Assume that the previous test does not reject the null hypothesis. Does this result provide strong evidence that the pull-off force is > 25 N. Explain?

The null and alternative hypothesis used to demonstrate

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that the pull-off force is below 25 newtons are

$H_0: \mu = 25N$.
$H_1: \mu < 25N$.

b. No. The truth or falsity of a particular hypothesis can never be known with certainty unless we can examine the entire popln.

4. A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a std devn. of 0.5 kg. The company wishes to test the hypothesis $H_0: \mu = 12$ against $H_1: \mu < 12$, using a random sample of 4 specimens.

a) What is the type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?

(b) Find β for the case where the true mean elongation is 11.25 kg.

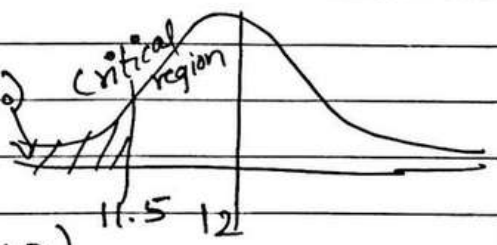
(c) Find β for the case where the true mean is 11.5 kgs?

a. The probability of making a type I error is

$\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

$\mu = 12$ kg $\quad \sigma = 0.5$ kg

$\bar{x} < 11.5$ kg $\rightarrow$ critical region



$$\alpha = P(\bar{x} < 11.5 \text{ when } \mu = 12)$$

$$= P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < \frac{11.5 - \mu}{\sigma/\sqrt{n}} \text{ when } \mu = 12\right)$$

$$= P\left(Z < \frac{11.5 - 12}{0.5/\sqrt{4}}\right) = P(Z < -2)$$

$\therefore$ from tables for $Z < -2$ prob $= 0.022750$.

$\therefore \alpha = 0.02275$.

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b. β = P(type II error) = P(fail to reject H_0 when H_0 is false)

$$\beta = P(\bar{x} > 11.5 \text{ when } \mu = 11.25)$$

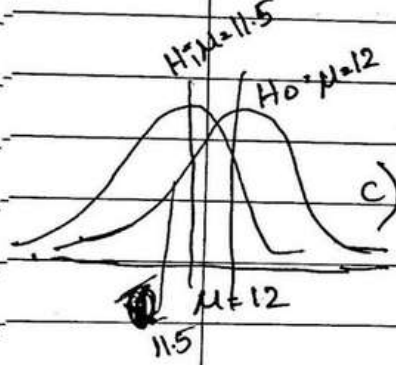
$$= P\left(\frac{\bar{x}-\mu}{\sigma/\sqrt{n}} > \frac{11.5-\mu}{\sigma/\sqrt{n}} \quad \mu = 11.25\right)$$

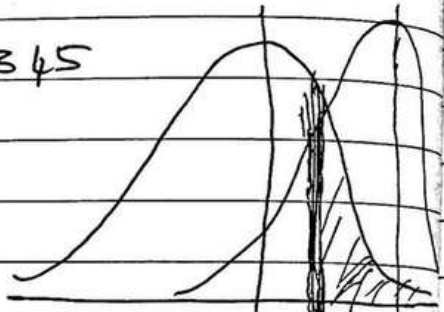
$$= P\left(\frac{\bar{x}-11.25}{0.5\sqrt{4}} > \frac{11.5-11.25}{0.5\sqrt{4}}\right)$$

$$= P(z > 1) = 1 - P(z \leq 1)$$

$$= 1 - 0.15865 = 1 - 0.841345$$

$$= 0.15865$$





c) when $\mu = 11.5$

$$\beta = P(\bar{x} > 11.5 \text{ when } \mu = 11.5)$$

$$= P\left(\frac{\bar{x}-11.5}{0.5/\sqrt{4}} > \frac{11.5-11.5}{0.5/\sqrt{4}}\right)$$

c).

$$= P(z > 0) = 1 - P(z < 0) = 1 - 0.5 = 0.5$$

5. Length L of certain speciality parts is normally distributed with variance of $50mm^2$. If a customer wants the mean length to be 100 mm but measures the avg length in a randomly selected sample as 103mm, what is the conclusion about the sample? Sample size, n = 50 parts.

$H_0: \mu = 100$ Vs. $H_1: \mu \neq 100$.

$$Z_0 = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{103-100}{\sqrt{50/50}} = \frac{3}{1} = 3.$$

Level of confidence → 95%.

Assuming $\alpha = 5\%$ $Z_{\alpha/2} = Z_{0.025} = 1.96$.

Since $Z_0 > Z_{\alpha/2}$, reject H_0 ⇒ Mean of parts ≠ 100

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The customer rejects the sample.

4 continued If the sample size $n=16$ & critical region same.

1. $\alpha = P(\bar{X} < 11.5 \text{ when } \mu = 12)$

$= P\left(Z < \frac{11.5-12}{0.5/\sqrt{16}}\right)$

$= P\left(Z < \frac{-0.5}{0.5/4}\right)$ $\quad P(Z<-4) = 0.000$.

2. $\beta = P(\bar{X} > 11.5 \text{ when } \mu = 11.25)$

$= P\left(Z > \frac{11.5-11.25}{0.5/\sqrt{16}}\right) = P\left(Z > \frac{-0.25}{0.5/4}\right)$

$= P(Z > -2) = 1 - P(Z < -2) = 0.022750$.

3. $\beta = P(\bar{X} > 11.5 \text{ when } \mu = 11.5)$

$= P\left(Z > \frac{11.5-11.5}{0.5/\sqrt{16}}\right) = 1 - P(Z<0) = \underline{0.5}$

continued → Find the boundary of the critical region if the type I error probability is

a) $\alpha = 0.01$ and $n=4$ c) $\alpha = 0.01$ & $n=16$.

b) $\alpha = 0.05$ & $n=4$

(a) $\alpha = 0.01$ & $n=4$.

$0.01 = P\left(Z \leq \frac{C-12}{0.5/\sqrt{4}}\right)$

Prob of $0.01 \Rightarrow Z = -2.33$.

$\therefore \frac{C-12}{0.5/\sqrt{4}} = -2.33 \Rightarrow C = 11.4175$

b) $\alpha = 0.05 \Rightarrow z = -1.65$.

$\therefore \frac{C-12}{0.5/\sqrt{4}} = -1.65 \Rightarrow C = 11.5875$

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c) $\alpha = 0.01$ & $n = 16$

$\Rightarrow z = -2.33$

$\therefore \frac{c - 12}{0.5/\sqrt{16}} = -2.33$

$\therefore c = -0.29125 + 12 = 11.70875$

6. The heat evolved in calories/gm of a cement mixture is approximately normally distributed. The mean is thought to be 100 & the std dev. is 2. We wish to test $H_0: \mu = 100$ versus $H_1: \mu \neq 100$ with a sample of $n = 9$ specimens.

a) If the acceptance region is defined as $98.5 \leq \bar{x} \leq 101.5$ find the type 1 error probability.

b) Find β for the case where the true mean heat evolved is 103.

c) Find β for the case where the true mean heat evolved is 105. Compare β in (b) and (c) and state reasons.

$H_0: \mu = 100 \quad H_1: \mu \neq 100 \quad n = 9$

(a) Prob of (type I error) $= \alpha$

Acceptance region is $98.5 \leq \bar{x} \leq 101.5$

$\alpha = P(\bar{x} < 98.5) + P(\bar{x} > 101.5)$

$= P\left(z < \frac{98.5 - 100}{2/\sqrt{9}}\right) + P\left(z > \frac{101.5 - 100}{2/\sqrt{9}}\right)$

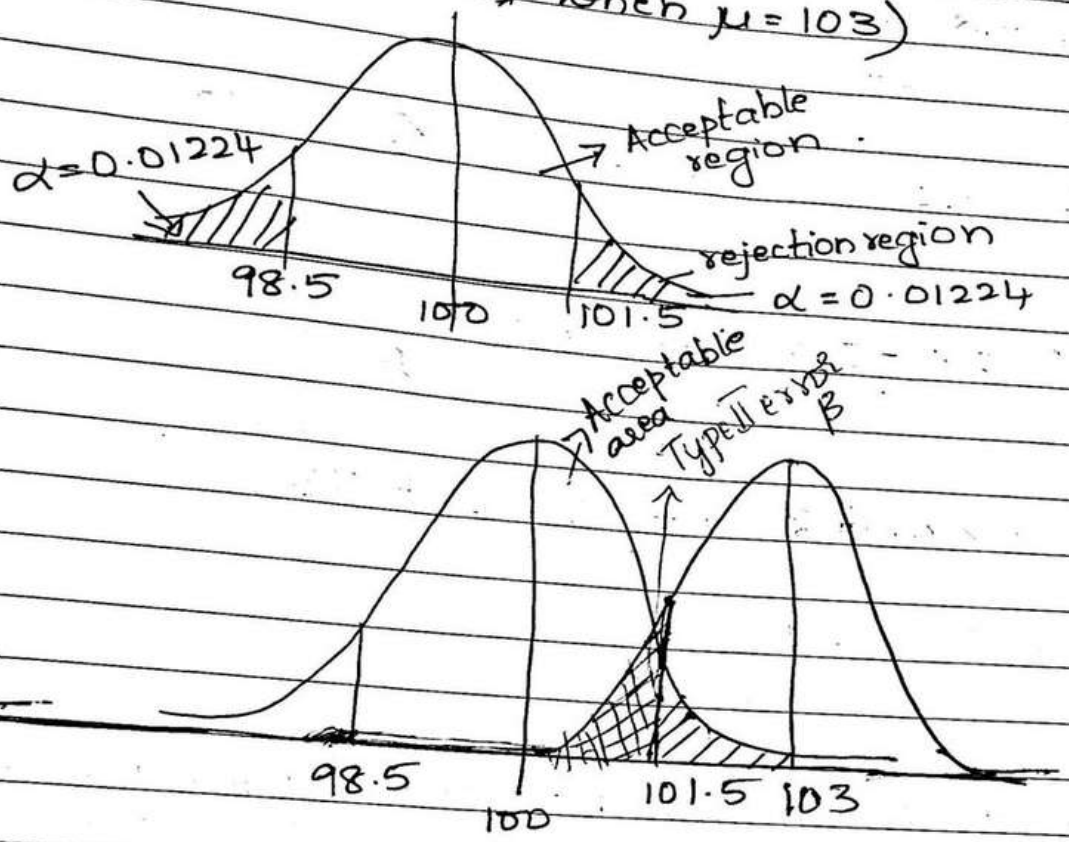
$= P\left(z < \frac{(-1.5) * 3}{2}\right) + P\left(z > \frac{4.5}{2}\right)$

$= 0.012224 + 1 - P(z < 2.25)$

$= 0.012224 + 1 - 0.987776$

$= 0.024448.$

b. $\beta = P(98.5 \leq \bar{x} \leq 101.5)$ when $\mu = 103$)



$= \left(P\left(\frac{98.5-\mu}{\sigma/\sqrt{n}} \leq z \leq \frac{101.5-\mu}{\sigma/\sqrt{n}}\right) \text{ when } \mu = 103\right)$

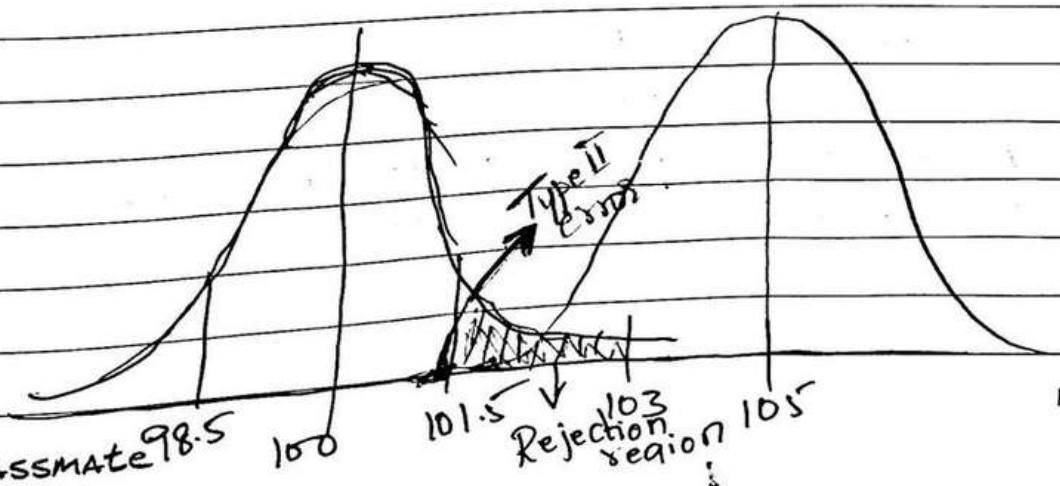
$= P\left(\frac{98.5-103}{2/\sqrt{9}} \leq \frac{\bar{x}-103}{\sigma/\sqrt{n}} \leq \frac{101.5-103}{2/\sqrt{9}}\right)$

$= P(-6.75 \leq z \leq -2.25)$

$= P(z \leq -2.25) - P(z \leq -6.75)$

$\boxed{\beta = 0.012224}$

c. When $\mu = 105$.



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$$\beta = P\left(\frac{98.5-\mu}{2/\sqrt{9}} \leq \frac{\bar{x}-\mu}{6/\sqrt{n}} \leq \frac{101.5-\mu}{2/\sqrt{9}} \text{ when } \mu=105\right) \quad (z)$$

$$= P\left(z < \frac{101.5-105}{(2/\sqrt{9})}\right) - P\left(z \leq \frac{98.5-105}{2/\sqrt{9}}\right)$$

$$= P\left(z < -\frac{3.5}{2/3}\right) - P\left(z \leq \frac{-6.5}{2/3}\right)$$

$$= P(z < -5.25) - P(z \leq -9.75)$$

$$\beta = 0 - 0 = \underline{0}$$

We can observe than when true mean is 103 there was 1% chance of failing to reject null hypothesis even though it is false, but when true mean is 105 we have to definitely ~~decreas~~ reject null hypothesis for the given Confidence Interval.

1. A manufacturer is interested in the output voltage of a power supply used in a PC. O/p voltage is assumed to be normally distributed with std devn. 0.25 volts & manufacturer wishes to Test $H_0: \mu = 5$ volts against $H_1: \mu \neq 5$ volts using $n = 8$ units.

a) The acceptance region is $4.85 \leq \bar{x} \leq 5.15$. Find the value of α.

b) Find the power of the test for detecting a true mean O/p voltage of 5.1 volts.

Soln $H_0: \mu = 5$ Vs $H_1: \mu \neq 5$ $n = 8$. $\sigma = 0.25$

a) $\alpha = P\left(\frac{4.85-\mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15-\mu}{0.25/\sqrt{8}}\right)$, $Z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$

$= P(Z \leq -1.70) + (1 - P(Z \leq 1.70))$

$= 0.044565 + (1 - 0.9554435)$

$= 0.044565 + 0.044565 = 0.089130$

$= 0.089130 \simeq 0.089$

b) Power $= 1 - \beta$.

$\mu = 5.1 \text{ Volts}$.

$$\beta = P\left(\frac{4.85 - \mu}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - \mu}{0.25/\sqrt{8}} \quad \mu = 5.1\right)$$

$$= P\left(\frac{4.85 - 5.1}{0.25/\sqrt{8}} \leq Z \leq \frac{5.15 - 5.10}{0.25/\sqrt{8}}\right)$$

$= P(-2.8284 \leq z \leq 0.5657)$

$= P(z \leq 0.5657) - P(z \leq -28284)$

$\beta = 0.71186$

$\therefore$ power of the test $= 1 - \beta$

$= 1 - 0.71186 = \underline{\underline{0.28814}}$

8. The proportion of residents in Phoenix favoring the building of toll roads to complete the freeway system is believed to be $p = 0.3$. If a random sample of 10 residents shows that 1 or fewer favor this proposal, we will conclude $p < 0.3$.

a) Find the probability of type 1 error if true proportion is $p = 0.3$.

b) Find the prob. of commiting a type II error with this procedure if $p = 0.2$

c) What is the power of this procedure if true proportion is $p = 0.2$?

Soln. $H_0: p = 0.3$ $H_1: p < 0.3$ $n = 10$.

Let X: no. of residents in favor of this proposal, then x is a random variable with binomial distn with parameters ~~n~~ $n = 10$ & p.

$\therefore P(X = x) = ({}^{n}C_x)\, p^x (1-p)^{10-x}$ for $x \in \{1, \ldots n\}$

(a) $\alpha = P(\bar{x} \leq 1 \text{ when } p = 0.3)$

$= P(\bar{x} = 0 \text{ when } p = 0.3) + P(\bar{x} = 1 \text{ when } p = 0.3)$

$= ({}^{10}C_0)\, 0.3^0 (1-0.3)^{10-0} + ({}^{10}C_1)\, 0.3^1 (1-0.3)^{10-1}$

$= 1 * 0.7^{10} + 10 * 0.3 * 0.7^9$

$\alpha = 0.1493$

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b) $\beta = P(\bar{x} > 1 \text{ when } p = 0.2)$

$= 1 - P(\bar{x} < 1 \text{ when } p = 0.2)$

$= 1 - P(\bar{x} = 0 \text{ when } p = 0.2) + P(\bar{x} = 1 \text{ when } p = 0.2)$

$= 1 - ({}^{10}C_0)\, 0.2^0 * (1-0.2)^{10-0} + ({}^{10}C_1) * 0.2^1 * (1-0.2)^9$

$= 1 - 1 * 0.8^{10} + 10 * 0.2 * 0.8^9$

$\beta = 0.6242$.

c. If the true proportion is $p = 0.2$, the power is

$1 - \beta = 1 - 0.6242 = 0.3758$

Tests on the mean of a Normal distribution, variance known.

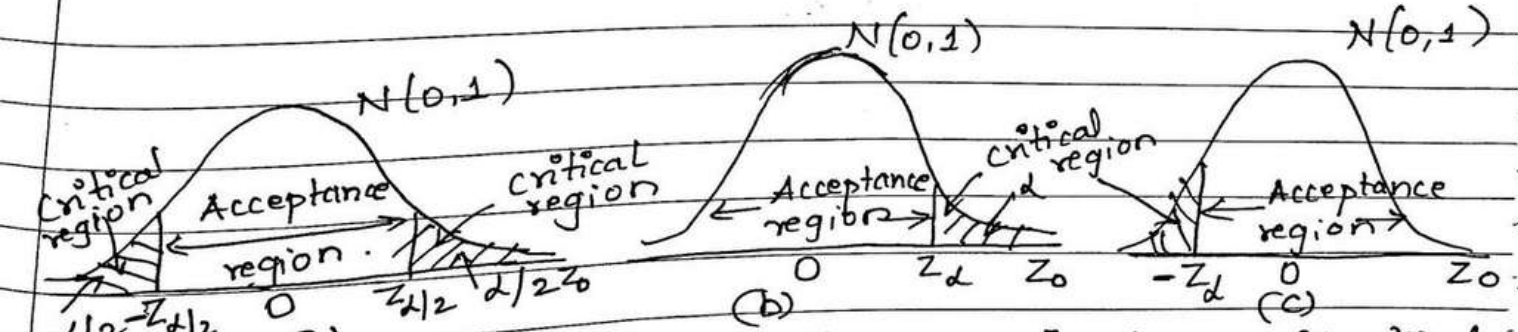
1. Hypothesis Tests on the Mean.

Null $H_0: \mu = \mu_0$.

Alternate $H_1: \mu \neq \mu_0$, $H_1: \mu > \mu_0$, $H_1: \mu < \mu_0$.

μ_0 is specified constant. We have a random sample $X_1, X_2, \ldots X_n$ from a normal population.

Test Statistic $Z_0 = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$.



The distribution of Z_0 when $H_0: \mu = \mu_0$ is true, with critical region for (a) the two sided alternative $H_1: \mu \neq \mu_0$ b) One sided alternative $H_1: \mu > \mu_0$ c) One sided alternative $H_1: \mu < \mu_0$,

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* For 2 sided test we should reject H_0 if the observed value of test statistic z_0 is either

$$z_0 > z_{\alpha/2} \text{ or } z_0 < -z_{\alpha/2} \quad -(1)$$

and we should fail to reject H_0 if

$$-z_{\alpha/2} \leq z_0 \leq z_{\alpha/2} \quad -(2).$$

The 2 inequalities in eqn (1) define the critical region or rejection region.

Eqn 2 define the acceptance region.

or can be written as.

Reject $H_0: \mu = \mu_0$ if either $\bar{x} > a$ or $\bar{x} < b$ where

$$a = \mu_0 + z_{\alpha/2}\sigma/\sqrt{n} \qquad b = \mu_0 - z_{\alpha/2}\sigma/\sqrt{n}.$$

* One sided alternative.

$H_0: \mu = \mu_0$ $H_1: \mu > \mu_0$

Rejection criteria $z_0 > z_{\alpha, n-1}$.

or $\bar{x} > z_{\alpha, n-1}$

* One sided alternative

$H_0: \mu = \mu_0$ $H_1: \mu < \mu_0$

Rejection criteria $z_0 < -z_\alpha$.

Pvalues

for the foregoing Normal distn test, P value is computed as foll.

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$$P = \begin{cases} 2[1-\phi(|z_0|)] & \text{for a two tailed test } H_0: \mu=\mu_0 \quad H_1: \mu \neq \mu_0 \\ 1-\phi(z_0) & \text{for upper tailed test } H_0: \mu=\mu_0 \quad H_1: \mu > \mu_0 \ (\mu > \mu_0) \\ \phi(z_0) & \text{for lower tailed test } H_0: \mu=\mu_0 \quad H_1: \mu < \mu_0 \end{cases}$$

Here $\phi(z)$ is the std normal cum. fn.

$\phi(z) = P(Z \leq z)$ where Z is $N(0,1)$

** The P-value is the smallest level of significance that would lead to rejection of the null hypothesis H_0 with the given data.

Prob. of a Type-II Error for a 2 sided Test on mean, Variance Known.

$$\beta = \phi\left(z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \phi\left(-z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

or if $\delta > 0$ $\quad \beta \simeq \phi\left(z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$

~~where $\delta = \mu - \mu_0$.~~

Sample size.

$$n \simeq \frac{(z_{\alpha/2} + z_\beta)^2 \sigma^2}{\delta^2} \quad \text{where } \delta = \mu - \mu_0.$$

Problems.

1. State the null & alternative hypothesis in each case

(a) A hypothesis test will be used to potentially provide evidence that the popln. mean is greater than 10.

(b) A hypothesis test will be used to potentially provide evidence that the population mean is not equal to 7.

c) A hypothesis test will be used to potentially provide evidence that the population mean is < 5.

a) $H_0: \mu = 10, \quad H_1: \mu > 10$.

b) $H_0: \mu = 7, \quad H_1: \mu \neq 7$

c) $H_0: \mu = 5, \quad H_1: \mu < 5$

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2. A hypothesis will be used to test that a population mean equals 10 against the alternative that the popul mean is greater than 10 with known variance 6. What the critical value for the test statistic Z_0 for the foll significance levels?

a) 0.01 b) 0.05 c) 0.10.

$H_0: \mu = 10$ Vs $H_1: \mu > 10$.

Test statistic $Z_0 = \frac{\bar{x} - \mu}{6/\sqrt{n}}$.

for 1 sided alternative hypothesis as in this case

$P = (1 - \phi(z_c)$

a) $\alpha = 0.01$ = significance level.

$\therefore P = Z_0 < z_c = 1 - 0.01 = 0.99$.

$\therefore$ critical values for test statistic Z_0 are

$Z_c = 2.33$

b. $\alpha = 0.05$

$\therefore P = Z_0 < z_c = 1 - 0.05 = 0.95$

$\therefore$ critical value for test statistic Z_0 are

$Z_c = 1.65$.

c) $\alpha = 0.10$

$P = Z_0 < Z_c = 1 - 0.1 = 0.9$.

$Z_c = 1.29$

3. A hypothesis will be used to test that a population mean equals 7 against the alternative that the popln mean does not equal 7 with known variance 6. What are the critical values for the test statistic Z_0 for the foll significance levels a) 0.01 b) 0.05 c) 0.10

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$H_0: \mu = 7$ $H_1: \mu \neq 7$.

$$z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}.$$

Pvalue $= P = 2[1 - \phi(|z_c|)]$

a. For $\alpha = 0.01$.

$$P(z_0 < |z_c|) = 1 - \frac{0.01}{2} = 0.995$$

Critical value of test statistic z_0 are

$$z_c = 2.58 \Rightarrow z_c = \pm 2.58$$

b. $\alpha = 0.05$, $P(z_0 < |z_c|) = 1 - \frac{0.05}{2} = 0.975$.

Critical value $z_c = 1.96$ $z_c = \pm 1.96$.

c. $\alpha = 0.1$ $P[z_0 < |z_c|) = 1 - \frac{0.1}{2} = 0.95$ $z_c = \pm 1.65$

4. For the hypothesis test $H_0: \mu = 7$ and $H_1: \mu \neq 7$ & variance known, calculate P-value for each of the foll test statistics.

a) $z_0 = 2.05$ b) $z_0 = -1.84$ c) $z_0 = 0.4$

Pvalue for 2sided test

$P = 2[1 - \phi[|z_0|]$ for

a) $z_0 = 2.05$ $P = 2[1 - \phi[2.05]] = 2[1 - 0.979818]$

$= 2*(0.020182) = 0.040364$

b) $z_0 = -1.84 = 2*[1 - 0.967116] = 2*[0.032334]$

$= 0.64668$

c) $z_0 = 0.4 = 2*[1 - 0.655422) = 2*[0.344578]$

$= 0.689156 \simeq 0.69$

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5. A manufacturer produces crankshafts for an automobile engine. The wear of the crankshaft after 100,000 miles (0.0001 inch) is of interest because it is likely to have an impact on warranty claims. A random sample of n=15 shafts is tested and $\bar{x} = 2.78$. It is known that $\sigma = 0.9$ and that wear is normally distributed.

(a) Test $H_0: \mu = 3$ versus $H_1: \mu \neq 3$ using $\alpha = 0.05$.

(b) What is power of this test if $\mu = 3.25$?

(c) What sample size would be required to detect a true mean of 3.75 if we wanted the power to be atleast 0.9?

$H_0: \mu = 3$ Vs $H_1: \mu \neq 3$. $\alpha = 0.05$.

$\bar{X} = 2.78$ $\sigma = 0.9$

$n = 15$.

(a) $Z_0 = \dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}}$.

$Z_0 = \dfrac{2.78 - 3}{0.9/\sqrt{15}} = -0.9467$

Two sided test $\therefore Z_{\alpha/2} = Z_{0.025} = 1.96 = \pm 1.96$

Rejection criteria for 2 sided test is

$Z_0 > Z_{\alpha/2}$ or $Z_0 < -Z_{\alpha/2}$.

Here $Z_0 = -0.95 > -1.96$ but $-0.95 < 1.96$.

$\therefore$ We fail to reject H_0

(b) $\beta = \phi\left(Z_{\alpha/2} - \dfrac{\delta\sqrt{n}}{\sigma}\right) - \phi\left(-Z_{\alpha/2} - \dfrac{\delta\sqrt{n}}{\sigma}\right)$

where $\delta = \mu - \mu_0$.

here $\mu = 3.25$ $\mu_0 = 3$ $\therefore \delta = 0.25$.

$\beta = \phi\left(1.96 - \dfrac{0.25 * \sqrt{15}}{0.9}\right) - \phi\left(-1.96 - \dfrac{0.25 * \sqrt{15}}{0.9}\right)$

$= \phi(0.88) - \phi(-3.04)$

$= 0.809387$

$\therefore$ Power $= 1 - \beta = 1 - 0.809359 = 0.19061$

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c. Sample size

$$n = \frac{(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{\delta^2}$$

$\mu = 3.75 \quad \delta = 3.75 - 3.0 = 0.75$

power $= 0.9 \quad \therefore \beta = 0.1$

$z_\beta = z_{0.1} = 1.28$

$$n = \frac{(1.96 + 1.28)^2 * 0.9^2}{0.75^2}$$

$$= 15.116$$

$\boxed{n \approx 16}$

6. The life in hours of a battery is known to be approx. normally distributed with std. devn $\sigma = 1.25$ hours. A random sample of 10 batteries has a mean life of $\bar{x} = 40.5$ hrs.

a) Is there evidence to support the claim that battery life exceeds 40 hrs? Use $\alpha = 0.05$

b) What is the P-value for the test in part (a)?

c) What is the β-error for the test in part (a) if the true mean life is 42 hours?

d) What sample size would be required to ensure that β does not exceed 0.10 if true mean life is 44 hrs?

e) Explain how you could answer the question in part (a) by calculating an appropriate confidence bound on life.

Soln Given $H_0: \mu = 40$ Vs $H_1: \mu > 40$.

$\alpha = 0.05 \quad n = 10 \quad \sigma = 1.25$

Test statistic $z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{40.5 - 40}{1.25/\sqrt{10}} = 1.26$

$z_\alpha = z_{0.05} = 1.65$

For 1 sided test $\mu > \mu_0 \quad z_0 > z_\alpha \rightarrow$ rejection criteria

Here $z_0 = 1.26$ & $z_\alpha = 1.65$

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Since $z_0 < z_\alpha$ we fail to reject null hypothesis or there is ~~sufficient~~ evidence to support the claim the battery life exceeds 40 hours

b. The P-value for this test is

$$P(z_\alpha > z_0) = 1 - \phi(z_0)$$
$$= 1 - \phi(1.26) = 1 - 0.896165$$
$$= 0.103835$$

c. $\beta = \phi\left(z_\alpha - \frac{\delta\sqrt{n}}{\sigma}\right)$ $\quad \delta = \mu - \mu_0$.

$\delta = 42 - 40 = 2$

$$\beta = \phi\left(1.65 - \frac{2\sqrt{10}}{1.25}\right) = \phi(-3.41)$$

$\beta = 0.000325$

d. $n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}$

here $\mu = 44$ $\delta = 4$ We want $\beta < 0.1$
$z_\beta = z_{0.1} = 1.29$ and

$$n = \frac{(1.65 + 1.29)^2 \; 1.25^2}{4^2} = 0.844$$

$n \cong 1$

e. A $(100 * (1-\alpha))\%$ lower bound on the mean is

$$\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}} \leq \mu$$

$\therefore$ 95% lower bound on the mean is

$$40.5 - 1.65 * \frac{1.25}{\sqrt{10}} \leq \mu.$$

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$39.85 \leq \mu$

$40 \notin (39.85, \mu)$ → provides evidence to support that battery life exceeds 40hrs

7. An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approx. normally distributed with $\sigma = 60$ psi. A random sample of 12 specimens has a mean tensile strength of $\bar{x} = 3450$ psi.

(a) Test the hypothesis that mean strength is 3500 psi. Use $\alpha = 0.01$.

b) What is the smallest level of significance at which you would be willing to reject the null hypothesis?

c) What is the β-error for the test in part (a) if the true mean is 3470?

d) Suppose that we wanted to reject the null hypothesis with probability at least 0.8 if mean strength $\mu = 3500$. What sample size should be used?

e) Explain how you could answer the question in part (a) with a 2 sided confidence interval on mean tensile strength.

Soln. Given $n = 12$ $\bar{X} = 3450$ psi $\sigma = 60$

$H_0: \mu = 3500$ $H_1: \mu \neq 3500$.

$$Z_0 = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{3450 - 3500}{60/\sqrt{12}} = -2.89.$$

$\alpha = 0.01 \Rightarrow Z_{\alpha/2} = Z_{0.005} = \pm 2.58$

Rejection criteria ~~$-Z_\alpha \leq Z_0 \leq Z_\alpha$~~ $Z_0 \leq -Z_\alpha$ or $Z_0 > Z_\alpha$

Since $Z_0 = -2.89 \leq -2.58$ we reject null Hypothesis

i.e $\mu = 3500$ is not true

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b) Smallest level of significance or Pvalue

$$\text{Pvalue} = 2[1-\phi(|z_0|)]$$
$$= 2[1-\phi(2.89)]$$
$$= 2[1-0.998074]$$
$$= 0.003852$$

c) Prob. of type II error.

$$\beta = \phi\left(z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \phi\left(-z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$$

$\delta = \mu - \mu_0$

$\mu_0 = 3500 \quad \mu = 3470$

$\delta = -30.$

$$\beta = \phi\left(2.58 - \frac{-30\sqrt{12}}{60}\right) - \left(-2.58 - \frac{-30\sqrt{12}}{60}\right)$$

$$= \phi(4.31) - \phi(-0.85)$$

$$= 1 - 0.197662. \quad = 0.802338.$$

d) Sample size

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}. \qquad \delta = \mu - \mu_0$$

$\mu = 3470 \quad \delta = -30 \qquad$ power $= 0.8 \therefore \beta = 0.2$

$z_\beta = z_{0.2} = 0.84$ and

$$n = \frac{(2.58 + 0.84)^2 * 60^2}{(-30)^2} = 46.78.$$

$n \simeq 47$

e) a $100(1-\alpha)\%$ 2 side confidence interval on mean

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

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$= 3450 - 2.58 * \frac{60}{\sqrt{12}} \leq \mu \leq 3450 + 2.58 * \frac{60}{\sqrt{12}}$

$3405.31 \leq \mu \leq 3494.67$

As 3500 is not in the interval, we reject H_0 i.e $\mu = 3500$ is not true for $\alpha = 0.01$.

8. A bearing used in an automotive appln is suppose to have a nominal inside diameter of 1.5 inches. A random sample of 25 bearings is selected & the avg inside dia of these bearings is 1.4975 inches. Bearing diameter is known to be normally distributed with std. devn. $\sigma = 0.01$ inch.

a) Test the hypotheses $H_0: \mu = 1.5$ vs $H_1: \mu \neq 15$ using $\alpha = 0.01$

b) What is the pvalue for the test

c) Compute the power of the test if the true mean dia is 1.495 inches.

d) What sample size would be required to detect a true mean dia as low as 1.495" if we wanted the power of test to be atleast 0.9?

e) Explain how (a) can be answered by constructing a 2 sided C.I on the mean dia.

Soln $n = 25$ $\bar{x} = 1.4975"$ $\sigma = 0.01"$,

$H_0: \mu = 1.5$ $H_1: \mu \neq 1.5$ $\alpha = 0.01$

$Z_0 = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1.4975 - 1.5}{0.01/\sqrt{25}} = -1.25.$

$Z_{\alpha/2} = Z_{0.005} = \pm 2.58.$

Rejection criteria $Z_0 < -Z_{\alpha/2}$ or $Z_0 > Z_{\alpha/2}$

Since $Z_0 > -Z_{\alpha/2}$ but $Z_0 < Z_{\alpha/2}$ we fail to reject H_0.

$\therefore H_0: \mu = 1.5$ is true.

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b. The P-value of this test

$$P\text{-value} = 2[1 - \phi(|z_0|)]$$
$$= 2[1 - \phi(1.25)]$$
$$= 2[1 - 0.894350] = 0.2113$$

c) Power = $1-\beta$.

β or prob of Type II error $= \phi\left(z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right) - \phi\left(-z_{\alpha/2} - \frac{\delta\sqrt{n}}{\sigma}\right)$

$\delta = \mu - \mu_0 \quad \delta = 1.495 - 1.5 = -0.005$.

$$\therefore \beta = \left(2.58 - \frac{-0.005 * \sqrt{25}}{0.01}\right) - \phi\left(-2.58 - \frac{-0.005 * \sqrt{25}}{0.01}\right)$$

$$= \phi(5.08) - \phi(-0.08)$$
$$= 1 - 0.468119 = 0.531881$$

Power of the test is $1-\beta = 1 - 0.531881 = 0.468119$

Power = 0.468119

d. $n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}$ $\quad \delta = 1.495 - 1.5 = -0.005$

Power $= 0.9 = 1-\beta \Rightarrow \beta = 0.1$

$z_{0.1} = 1.28$ and

$$n = \frac{(2.58 + 1.28)^2 * 0.01^2}{(-0.005)^2} = 59.6$$

$n \simeq 60$

e A $(100 * (1-\alpha))\%$ 2 sided CI on the mean is

$$\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$1.4975 - \frac{2.58 * 0.01}{\sqrt{25}} \leq \mu \leq 1.4975 + \frac{2.58 * 0.01}{\sqrt{25}}$$

$$= 1.49234 \leq \mu \leq 1.50266$$

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Since 1.5 is in the interval [1.49234, 1.50266] we can conclude $H_0: \mu = 1.5$ is true & cannot be rejected.

Tests on the variance and standard deviation of a normal distribution

Hypothesis tests on the variance

$$H_0: \sigma^2 = \sigma_0^2$$
$$H_1: \sigma^2 \neq \sigma_0^2$$

Test statistic $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}$.

Null hypothesis would be rejected if

$$\chi_0^2 > \chi^2_{\alpha/2, n-1} \text{ or if } \chi_0^2 < \chi^2_{1-\alpha/2, n-1}$$

Rejection criteria

$H_0: \sigma^2 = \sigma_0^2$	
$H_1: \sigma^2 \neq \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha/2, n-1}$ or $\chi_0^2 < -\chi^2_{\alpha/2, n-1}$
$H_1: \sigma^2 > \sigma_0^2$	$\chi_0^2 > \chi^2_{\alpha, n-1}$
$H_1: \sigma^2 < \sigma_0^2$	$\chi_0^2 < -\chi^2_{\alpha, n-1}$

Problem

1. An automated filling m/c is used to fill bottles with liquid detergent. A random sample of 20 bottles results in a sample variance of fill volume of $s^2 = 0.0153$ (fluid ounces)^2. If the variance of fill volume exceeds 0.01 (" ")^2, an unacceptable proportion of bottles will be underfilled or overfilled. Is there evidence in the sample data to suggest that the manufacturer has a problem with underfilled or overfilled bottles? Use $\alpha = 0.05$ & assume that fill volume has a normal distribution.

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Using eight step procedure.

1. The parameter of interest is population variance σ^2
2. $H_0: \sigma^2 = 0.01$.
3. $H_1: \sigma^2 > 0.01$,
4. $\alpha = 0.05$.
5. Test statistic is

$$\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

6. Reject H_0 if $\chi_0^2 > \chi^2_{0.05,19} = 30.14$.

7. $\chi_0^2 = \frac{19 * 0.0153}{0.01} = 29.07$

8. Conclusions: Since $\chi_0^2 = 29.07 < \chi^2_{0.05,19} = 30.14$, we conclude that there is no strong evidence that the variance of fill volume exceeds $0.01 (\text{fluid ounces})^2$.

2. Consider the test of $H_0: \sigma^2 = 7$ against $H_1: \sigma^2 \neq 7$. What are the critical values for the test statistic χ_0^2 for the foll significance levels & sample sizes.

(a) $\alpha = 0.01$ & $n = 20$

(b) $\alpha = 0.05$ & $n = 12$.

(c) $\alpha = 0.10$ & $n = 15$.

(a) $\alpha = 0.01$ $n = 20$ $\alpha/2 = 0.005$. $\alpha/2 = 1 - 0.005 = 0.995$
critical values 6.84 & 38.58 for $n = 19$.

(b) $\alpha = 0.05$ $n = 12$ $\alpha/2 = 0.0025$ $\alpha/2 = 1 - 0.0025 = 0.975$
critical values 3.82 and 21.92. for $n = 11$

(c) $\alpha = 0.10$ $n = 15$ $\alpha/2 = 0.05$ $\alpha/2 = 0.95$
critical values 6.57 and 23.68.

2. Consider the test of $H_0: \sigma^2 = 10$ against $H_1: \sigma^2 > 10$ What are the critical value for the test statistic χ_0^2 for the foll significance levels & sample sizes?

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a) $\alpha = 0.01$ & $n = 20$

b) $\alpha = 0.05$ & $n = 12$.

c) $\alpha = 0.10$ & $n = 15$.

a) $\alpha = 0.01$ $n-1 = 19$. critical value $\chi^2_{0\,\alpha,n-1} = \chi^2_{0.01,19}$

$= 36.19$.

b) $\alpha = 0.05$ & $n = 12$ critical value $\chi^2_{0,\alpha,n-1} = \chi^2_{0.05,11} = 19.68$.

c) $\alpha = 0.10$ & $n = 15$ critical value $\chi^2_{\alpha,n-1} = \chi^2_{0.1,14} = \underline{21.06}$

3. $H_0: \sigma^2 \geq 5$ against $H_1: \sigma^2 < 5$ find critical values.

a) $\alpha = 1 - 0.01 = 0.99$ $n-1 = 19$ $\chi^2_{0.99,19} = 7.63$.

b) $\alpha = 1 - 0.05 = 0.95$, $n-1 = 11$ $\chi^2_{0.95,11} = 4.57$.

c) $\alpha = 1 - 0.10 = 0.90$ $n-1 = 14$ $\chi^2_{0.90,14} = 7.79$

4. The data from Medicine and Science in Sports and Exercise considered ice hockey player performance after electrostimulation training. In summary there were 17 players & the sample std dev of performance was 0.09 seconds.

a) Is there strong evidence to conclude that the std. dev of performance time exceeds the historical value of 0.75 secs? Use $\alpha = 0.05$. Find the p-value for this test.

b) Discuss how part (a) could be answered by constructing a 95% one sided confidence interval for σ.

Soln $n = 17$ $s = 0.09$ secs.

$H_0: \sigma^2 = 0.75$ $H_1: \sigma > 0.75$. $\alpha = 0.05$.

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Test statistic $\chi_o^2 = \frac{(n-1)s^2}{\sigma_o^2}$

$$\chi_o^2 = \frac{16 * (0.09)^2}{(0.75)^2} = 0.2304$$

$$\chi^2_{critical} = \chi^2_{0.05, n-1} = \chi^2_{0.05, 16} = 26.30$$

Since $\chi_o^2 < \chi^2_{critical}$ i.e $0.2304 < 26.30$ do not reject H_0 or fail to reject H_0. $\therefore \sigma^2$ may be 0.5625 or $\sigma = 0.75$.

P-value — Probability above $\chi_o^2 > 0.995$ (Based on Chi Square calculated value, check the table for which α value is nearer & pvalue for $H_1: \sigma > \sigma_o$, will be > that α value).

b. ~~95%~~ $100(1-\alpha)$ ~~$\alpha = 0.95$~~. $n = 16$.

$100(1-\alpha)\%$ $\alpha = 0.05$.

$\frac{(n-1)s^2}{\chi^2_{\alpha, n-1}} \leq \sigma^2$

~~$\chi^2_{0.95, 16} = 7.96$~~

$\chi^2_{0.05, 16} = 26.30$.

$= \frac{(17-1)\,0.09^2}{\text{~~7.96~~ } 26.30} \leq \sigma^2$

$= \boxed{0.004 \leq \sigma^2}$ or $\boxed{0.07 \leq \sigma}$.

5(i) What are the hypothesis statements and the statistic to be used in a 2 sided test of variance of a population? What are the assumptions to use this statistic & what is the rejection region for this test?

(ii) What is the value x such that a chisquare random variable with 15 degrees of freedom ~~taking on any value be~~ takes on at the most a value of x with 95% likelihood?

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$H_0: \sigma^2 = \sigma_0^2$ Vs $H_1: \sigma^2 \neq \sigma_0^2$ for 1 sided $H_1: \sigma^2 < \sigma_0^2$ $H_1: \sigma^2 > \sigma_0^2$

Statistic $\chi_0^2 = \frac{(n-1)s^2}{\sigma^2}$

Rejection Region χ_0^2 outside $\left(\chi^2_{\frac{1-\alpha}{2}, n-1}, \chi^2_{\alpha/2, n-1}\right)$

or $\chi_0^2 > \chi^2_{\alpha, n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha, n-1}$

Assumptions: The populations parameter are normal distributed.

(ii) $P(\chi^2_{15} < x) = 0.95 \Rightarrow P(\chi^2_{15} > x) = 0.05$.

$\Rightarrow x = \chi^2_{0.05, 15}$ → degrees of freedom

$\boxed{x = 25}$

6. Test at 10% significance if the variance of population is greater than 10, with the help of sample data, 24, 22, 25, 23, 20, 23, 21, 22, 25, 22. How can you answer this question with the help of confidence intervals?

$S^2 = \sum(x_i - \bar{x})^2/n-1$. $\bar{x} = 22.7$

$S^2 = 2.6778$

$\boxed{S = 1.63639}$

$H_0: \sigma^2 = 10$ Vs $H_1: \sigma^2 > 10$

$\alpha = 0.1$ $S = 1.63639$.

Test Statistic $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2}$

$= \frac{9 * (1.6364)^2}{10} = 2.4$

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$H_0: \sigma^2 = \sigma_0^2$ Vs $H_1: \sigma^2 \neq \sigma_0^2$ for 1 sided $H_1: \sigma^2 < \sigma_0^2$ $H_1: \sigma^2 > \sigma_0^2$

Statistic $\chi_0^2 = \frac{(n-1)s^2}{\sigma^2}$

Rejection Region χ_0^2 outside $\left(\chi^2_{\frac{1-\alpha}{2}, n-1}, \chi^2_{\alpha/2, n-1}\right)$

or $\chi_0^2 > \chi^2_{\alpha, n-1}$ or $\chi_0^2 < \chi^2_{1-\alpha, n-1}$

Assumptions: The populations parameter are normal distributed.

(ii) $P(\chi^2_{15} < x) = 0.95 \Rightarrow P(\chi^2_{15} \geq x) = 0.05$

$\Rightarrow x = \chi^2_{0.05, 15}$ → degrees of freedom

$\boxed{x = 25}$

6. Test at 10% significance if the variance of population is greater than 10, with the help of sample data, 24, 22, 25, 23, 20, 23, 21, 22, 25, 22. How can you answer this question with the help of confidence intervals?

$S^2 = \sum(x_i - \bar{x})^2/n-1$ $\quad \bar{x} = 22.7$

$S^2 = 2.6778$

$\boxed{S = 1.63639}$

$H_0: \sigma^2 = 10$ Vs $H_1: \sigma^2 > 10$

$\alpha = 0.1 \quad S = 1.63639.$

Test Statistic $\chi_0^2 = \frac{(n-1)S^2}{\sigma_0^2}$

$= \frac{9 * (1.63639)^2}{10} = 2.4$

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$\chi^2_{0.1,9}$ = from tables = 14.68.

Since $\chi_0^2 < \chi^2_{0.1,9}$ do not reject H_0
conclude that σ^2 may be = 10.

(b) 90% lower limit of σ^2:
$100(1-\alpha)\%$

$$\sigma^2 \geq \frac{(n-1)s^2}{\chi^2_{\alpha,n-1}} = \frac{9 * 1.6364^2}{14.68} = 1.642$$

The intervals suggest that σ^2 may be 10.

7. A research engineer for a tyre manufacturer is investigating tyre life for a new rubber compound and has built 16 tyres and tested them to end-of life in a road test. The sample mean and standard deviation are 60139.7 and 3645.94 kms.
i) Can you conclude, using $\alpha = 0.05$ that the std devn of tyre life exceeds 200 kms? State any necessary assumptions about the underlying distribution of the data
ii) Answer the question (i) by constructing appropriate confidence interval.

Soln.

Given $\bar{X} = 60139.7$ kms $s = 3645.94$ kms.
$n = 16$ $\alpha = 0.05$

$H_0: \sigma = 200$ $H_1: \sigma > 200$.

Test Statistic $\chi_0^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15 * 3645.94^2}{200^2}$

$= 4984.8 \approx \underline{4985}$

$\chi^2_{0.05,15} = 25$.

Since $\chi_0^2 > \chi^2_{0.05,15}$ we reject H_0

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$\therefore \sigma > 200.$

Assumption: The tyre life is normally distributed.

ii) $100(1-\alpha)\%$ CI (lower bound) = 95% lower bound CI on σ

$$= \sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}}} \leq \sigma$$

or $$\sigma \geq \sqrt{\frac{(n-1)s^2}{\chi^2_{\alpha,n-1}}} = \sqrt{\frac{15 \times 3645.94^2}{25}}$$

$$\sigma \geq 2824.13$$

Hence conclude that σ is far greater than 200.

Tests on the mean of a normal distribution, Variance unknown

Hypothesis Tests on the mean

If $X_1, X_2, \ldots X_n$ is a random sample from a normal distribution with mean μ and variance σ^2, the random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t distribution with $n-1$ degrees of freedom.

The gen. appearance of the 't' distribution is similar to the std. normal distribution in that both distributions are symmetric and unimodal & the max ordinate value is reached when the mean $\mu = 0$. However the t distribution has heavier tails than the normal, i.e it has more probability in the tails than the normal distribution.

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$H_0: \mu = \mu_0$

$H_1: \mu \neq \mu_0$

Test statistic

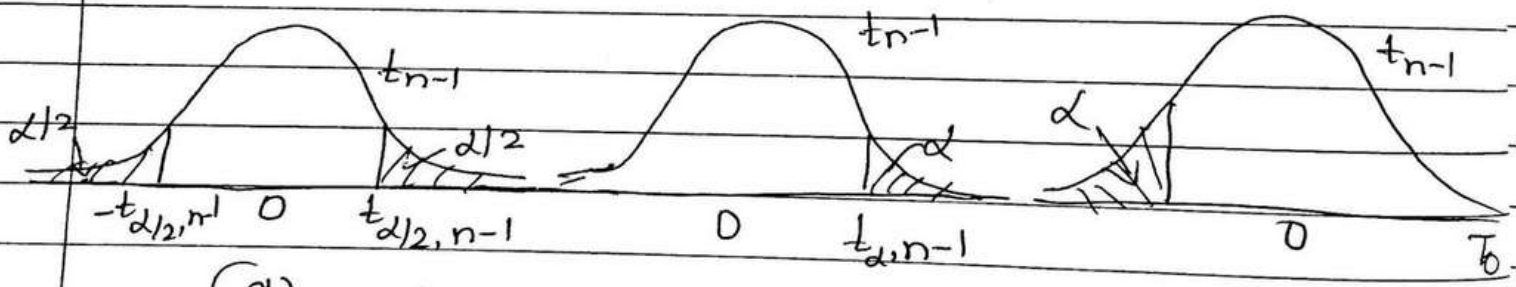
$$T_0 = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

If null hypothesis is true, T_0 has a 't' dist^n with n-1 degrees of freedom. We would use the 't' % points $-t_{\alpha/2, n-1}$ & $t_{\alpha/2, n-1}$ as boundaries of critical region

Null hypothesis $H_0: \mu = \mu_0$

Test Statistic $T_0 = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$.

Alternate hypothesis	Rejection criteria
$H_1: \mu \neq \mu_0$	$t_0 > t_{\alpha/2, n-1}$ or $t_0 < -t_{\alpha/2, n-1}$
$H_1: \mu > \mu_0$	$t_0 > t_{\alpha, n-1}$
$H_1: \mu < \mu_0$	$t_0 < -t_{\alpha, n-1}$



(a)

$H_0: \mu = \mu_0$ $H_1: \mu \neq \mu_0$ $H_1: \mu > \mu_0$ $H_1: \mu < \mu_0$.

Pvalue for a T-test

The P-value for at-test is smallest level of significance at which the null hypothesis would be rejected. i.e it is the tail area beyond the value of test statistic to for one sided test or twice this area for a 2 sided test.

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Computation of exact P-value directly from table is usually impossible. However it is easy to find upper & lower bounds on the P-value from this table.

Consider the foll critical values for t-test based on 14 Dof

Critical value	0.258	0.692	1.345	1.761	2.145	2.624	2.977
Tail area	0.4	0.25	0.1	0.05	0.025	0.01	0.005

If in a problem if $t_0 = 2.72$, this is lying in b/w values 2.624 & 2.977, therefore Pvalue must be b/w 0.01 & 0.005. These are effectively the upper & lower bounds on the P-value.

Problems.

1. A hypothesis will be used to test that a population mean equals 7 against the alternative that the population mean does not equal 7 with unknown variance 5. What are the critical values for the test statistic T_0 for the foll significance levels & sample sizes?

a) $\alpha = 0.01$ & $n = 20$

b) $\alpha = 0.05$ & $n = 12$

c) $\alpha = 0.10$ & $n = 15$.

$H_0: \mu = 7 \quad H_1: \mu \neq 7$.

a) $\alpha = 0.01 \quad \alpha/2 = 0.005 \quad t_{\alpha/2, n-1} = t_{0.005, 19} = 2.861$

$-t_{-\alpha/2, n-1} = -t_{0.005, 19} = -2.861$

$-2.861 \leq t_0 \leq 2.861$.

b) $\alpha = 0.05 \quad n = 12 \quad \alpha/2 = 0.025 \quad t_{\alpha/2, n-1} = 2.201$

$-2.201 \leq t_0 \leq 2.201$

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c) $\alpha = 0.1$ $n = 15$

$\alpha/2 = 0.05$ $n-1 = 14$ $t_{0.05,14} = 1.761$.

$-1.761 \leq t_0 \leq 1.761$

2. $H_0: \mu = 10$ $H_1: \mu > 10$ variance 6.

Find critical value a) $\alpha = 0.01$ & $n = 20$

b) $\alpha = 0.05$ & $n = 12$

c) $\alpha = 0.10$ & $n = 15$.

a) $\alpha = 0.01$ $n = 20$ $t_{\alpha, n-1} = t_{0.01, 19} = 2.539$

critical value

$\therefore t_0 > 2.539$.

b) $\alpha = 0.05$ $n = 12$ $t_{0.05, 11} = 1.796$.

$t_0 > 1.796$

c) $\alpha = 0.1$ $n = 15$ $t_{0.1, 14} = 1.345$.

Critical value > 1.345.

3. $H_0: \mu = 5$ $H_1: \mu \neq 5$. Variance 6.

a) $\alpha = 0.01$ $n = 20$ b) $\alpha = 0.05$ $n = 12$ c) $\alpha = 0.1$ & $n = 15$.

Ans. 1) Critical value < 2.539.

2) " " < 1.796

3) " " < 1.345.

4. For the hypothesis test $H_0: \mu = 7$ against $H_1: \mu \neq 7$ with variance unknown and $n = 20$, approximate the P-value for each of the foll test statistics.

a) $t_0 = 2.05$ b) $t_0 = -1.84$ c) $t_0 = 0.4$

a) $t_0 = 2.05$ for $n = 19$ from tables $t_0 = 2.05$ is lying b/w 1.729 & 2.093 ie for α b/w 0.05 & 0.025 ie $\frac{\alpha}{2} = 0.05$ & $\frac{\alpha}{2} = 0.025$ or $\alpha = 0.1$ & 0.05

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$\therefore 0.05 \leq p \leq 0.1$

b) $t_0 = -1.84$ is also lying $\frac{\alpha}{2} = 0.05$ & $\frac{\alpha}{2} = 0.001$

$\therefore 0.05 \leq p \leq 0.1$

c) $t_0 = 0.4$ $\frac{\alpha}{2} = 0.40$ to $\frac{\alpha}{2} = 0.25$

$0.5 \leq p \leq 0.8$

5. For the hypothesis test $H_0: \mu = 5$ against $H_1: \mu < 5$ with variance unknown & $n = 12$, approx. the pvalue for each of the foll. test statistics.

a) $t_0 = 2.05$ b) $t_0 = -1.84$ c) $t_0 = 0.4$

a) $n = 11$ $t_0 = 2.05 \rightarrow$ $\overset{\alpha}{0.05}$ to 0.0025

i.e. $0.95 \leq p \leq 0.975$

b) $t_0 = -1.84 \rightarrow 0.05$ to 0.025

$0.025 \leq p \leq 0.05$.

c) $t_0 = 0.4 \rightarrow 0.40$ to $0.25 \rightarrow \alpha$

$1 - \alpha = 0.6$ to 0.75

$0.6 \leq p \leq 0.75$.

6. The time in hours to install & commission a m/c is as follows for a sample of 10 machines chosen at random:

192, 201, 190, 183, 185, 176, 191, 214, 220, 205

i If the above data is assumed to come from a normal population can you conclude that the population mean is greater than 205 hrs at 5% significance level?

ii Derive an interval estimate to answer the same question.

Solution: $H_0: \mu = 205$ $H_1: \mu > 205$ hrs $\alpha = 0.05$

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Test statistic $t_0 = \frac{\bar{X} - \mu}{S/\sqrt{n}}$.

$\bar{X} = 195.7$.

$S^2 = \frac{1772.1}{10-1} = 196$

$\boxed{S = 14.032}$

$$t_0 = \frac{195.7 - 205}{14.0321/\sqrt{10}} = -2.0959.$$

$t_{\alpha, n-1} = t_{0.05, 9} = 1.833$

Since $t_0 < t_{\alpha, n-1}$ do not reject H_0

μ may be 205, not >205 hrs.

ii - Interval Estimate

Two sided. $= \bar{X} \pm t_{\alpha/2} \frac{S}{\sqrt{n}} = 195.7 \pm 2.262 * \frac{14.0321}{\sqrt{10}}$

$= (185.66, 205.74)$

Lower bound: $\mu \geq \bar{x} - t_\alpha \frac{S}{\sqrt{n}}$

$= 195.7 - 1.833 * \frac{14.052}{\sqrt{10}}$

$= 187.57$.

Both 2 sided & the lower bound CI include 205hrs
∴ we cannot reject claim that $\mu = 205$ hrs.

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7. A television documentary on overrating weight claimed that Americans are about 10 pounds overweigh on average. To test the claim, 18 randomly selected individuals were examined; their avg excess weight was found to be 12.4 pounds and the sample std. deviation was 2.7 pounds. At a significance level of 0.01, is there any reason to doubt the validity of the claimed 10-pound value.

$\bar{x} = 12.4$, $s = 2.7$ $\alpha = 0.01$ $n = 18$.

H_0: $\mu = 10$ pounds Vs H_1: $\mu > 10$ lb.

$$t_0 = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{12.4 - 10}{2.7/\sqrt{18}} = 3.77.$$

$$t_{\alpha, n-1} = t_{0.01, 17} = 2.567.$$

Since $t_0 > t_{\alpha, n-1}$ reject H_0, Data shows that Americans are overweight by >10 pounds average, & the claim of just 10lb overweight is not true.

8. State the probability of a t-random variable with 25 df taking on any value between 2.06 & 3.45.

$$P(2.06 < T_{25} < 3.45) = P(T > 2.06) - P(T > 3.45)$$
$$= 0.025 - 0.001 = 0.024$$

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TESTING FOR GOODNESS OF FIT.

Hypothesis testing procedures can be of 2 types, problems in which the probability distribution is known and the hypotheses involves the parameters of the distribution. Another kind of hypothesis is we do not know the underlying distribution of the population, and we wish to test the hypothesis that a particular distribution will be satisfactory as a population model.

The goodness of fit test procedure requires a random sample of size n from the population whose probability distribution is unknown. These 'n' observations are arranged in a frequency histogram having k bins or class intervals. Let O_i be the observed frequency in the ith class interval. From the hypothesized prob. distn, we compute the expected frequency in the ith class interval. denoted as E_i. The test statistic is:

$$\chi_0^2 = \sum_{i=1}^{k} (O_i - E_i)^2 / E_i$$

It can be shown that, if population follows the hypothesized distn, χ_0^2 has approximately, a chi-square distn with $k-p-1$ degrees of freedom, where p, represents the number of parameters of the hypothesized distribution estimated by sample statistics.

This approx improves as 'n' increases.

We would reject the hypothesis that the distn of the population is the hypothesized distn if the calculated value of the test statistic

$$\chi_0^2 > \chi^2_{\alpha, k-p-1}$$

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If the expected frequency are too small, the test statistic χ_0^2 will not reflect the departure of observed from expected, but only the small magnitude of expected frequencies. There is no gen. agreement regarding the minimum value of expected frequencies, but values of 3, 4 & 5 are widely used as minimal. If an expected frequency is too small, it can be combined with the expected frequency in an adjacent class interval. The corresponding observed frequencies would then also be combined & k would be reduced by 1. CI are not required to be of equal width

Problems

1. The no of defects in printed circuit boards is hypothesized to follow a Poisson distribution. A random sample of n=60 printed circuit boards has been collected, & the following number of defects observed. Perform a Test of hypothesis.

No of defects	Observed frequency
0	32
1	15
2	9
3	4

Soln

1. The variable of interest is the form of the distribution of defects in printed circuit boards.
2. H_0: The form of the distribution of defects is Poisson.
3. H_1: The form of the distribution of defects is not Poisson.
4. $\alpha = 0.05$.
5. Test statistic χ_0^2

Mean value of distn is not known. So to estimate it is the sample average.

$$= \frac{32*0 + 1*15 + 2*9 + 3*4}{60} = 0.75.$$

$\therefore \lambda = 0.75$.

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Computations.

7. We compute p_i, the theoretical, hypothesized probability associated with ith class interval.
Since i.e CI corresponds to no of defects, we may find p_i as follows

$$P_1 = P(X=0) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-0.75}\,0.75^0}{0!} = 0.472$$

$$P_2 = P(X=1) = \frac{e^{-0.75}\,0.75^1}{1!} = 0.354$$

$$P_3 = P(X=2) = \frac{e^{-0.75}\,0.75^2}{2!} = 0.133$$

$$P_4 = P(X \geq 3) = 1 - (p_1 + p_2 + p_3) = 0.041$$

Expected frequency are computed by multiplying samplesize $n = 60 \times$ probability i.e $E_i = n * p_i$

No. of defects	Prob.	E_i.	O_i	$(O_i - E_i)^2 / E_i$.
0	0.472	28.32	32	0.4781
1	0.354	21.24	15	1.833
2	0.133	7.8 } 10.44	9 } 13	0.6277
3 (or more)	0.041	2.46 }	4 }	

$$\chi^2_{cal} = 2.94$$

6. $\chi^2_{tab} = \chi^2_{0.05, k-p-1}$ $\quad p = 1$, $\because$ mean is estimated
$k = 3$ CI after pooling

$\chi^2_{0.05,1} = 3.84$.

8. Since $\chi^2_0 < \chi^2_{0.05,1}$, we fail to reject null hypothesis. that the distn. of defects in printed circuit boards is Poisson.

2. X, the number of reported safety incidents per day in a prod shop floor is recorded for 90 days as follows.

x_i no. accidents.	0	1	2	3	≥ 4
No. of days with x_i accident	54	23	9	3	1

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Does the data follow a Poisson distribution? Test at 5% significance level.

1. Point of interest → Data of accidents & their distn.
2. Ho: Data follows a Poisson distribution.
3. H_1: Data does not follow Poisson distn
4. $\alpha = 0.05$.
5. Test statistic $\chi_0^2 = \frac{(O_i - E_i)^2}{E_i}$.
6. Computations -

$$\lambda = \frac{0*54 + 1*23 + 2*9 + 3*3 + 4*1}{90} = 0.6$$

$$P_0 = P(X=0) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-0.6}\,0.6^0}{0!} = 0.5488$$

$P_1 = P(X=1) = 0.3293$

$P_2 = P(X=2) = 0.0988$.

$P_3 = P(X=3) = 0.0198$

$P_4 = P(X \geq 4) = 1 - 0.5488 - 0.3293 - 0.0988 - 0.0198$
$= 0.0033$

X_i	O_i	P_i	E_i		$(O_i - E_i)^2/E_i$
0	54	0.5488	49.39		0.4305
1	23	0.3293	29.64		1.4875
2	9	0.0988	8.89	10.97	0.3757
3	3	0.0198	1.79		
≥4	1	0.0033	0.3		

$\chi_0^2 = 2.2935$

$\chi^2_{tab} = \chi^2_{3-1-1,\,0.05} = 3.84$

7. Rejection criteria $\chi_0^2 < \chi^2_{tab}$ → Null hypothesis is not rejected

Data may be Poisson.

3. The production of radar component is checked by sample of four. The following table shows the number of defective found in 300 samples

No. of defectives	0	1	2	3	4
No of samples	90	130	58	20	2

Fit the binomial distn. for the data and test the goodness of fit at level of significance of 0.05.

Soln $\mu = np = 0*90 + 1*130 + 2*58 + 3*20 + 4*2/300$.

$= 314/300 = 1.047$.

Mean $= np = 1.047 = 4p \therefore p = 1.047/4 = 0.262$.

Value(X)	Prob.	Observed frequency	Expected	$\chi^2 = \frac{(O_i - E_i)^2}{E_i}$
0	0.2966	90	88.98	0.0117
1	0.4212	130	126.36	0.1049
2	0.2243	58	67.29	1.2826
3	0.0529	20 } 22	15.87 } 17.28	1.2867
4	0.0047	2 }	1.41 }	
			$\Sigma\chi^2$	2.6859

$P(X=0) = {}^nC_x p^x (1-p)^{n-x}$.

$= {}^4C_x * 0.262^x * 0.738^{4-x}$

$= {}^4C_0 * 0.262^0 * 0.738^4 = 0.2966$

$\chi^2_{tab} = \chi^2_{0.05, n-k-1} = \chi^2_{0.05, 4-1-1} = \chi^2_{0.05, 2} = 5.99$.

Since $\chi^2_{cal} = 2.6859 < \chi^2_{tab} = 5.99$ we fail to reject H_0. We can conclude the distribution is binomial

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with $n=4$ & $p=0.262$ at $\alpha=0.05$.

4 Nadir is testing an octahedral die to see if it is biased. The results are given as.

Score	1	2	3	4	5	6	7	8
Frequency	7	10	11	9	12	10	14	7

Test the hypothesis that the die is fair.

Soln

H_0: Die is fair.
H_1: Die is not fair.

Probability of occurring $= \frac{1}{8}$

It is uniform distribution & all dies side have equal probability.

$\alpha = 0.05$.

Degrees of freedom $\nu = n-k-1$
no parameter estimated $\therefore k=0$.
$\nu = 8-1=7$.

$\chi^2_{tab} = \chi^2_{0.05,7} = 14.067$.

Expected value $= \frac{1}{8}(7+10+11+9+12+10+14+7) = 10$.

Score	Frequency	Ei	$(O_i-E_i)^2/E_i$
1	7	10	0.9
2	10	10	0
3	11	10	0.1
4	9	10	0.1
5	12	10	0.4
6	10	10	0
7	14	10	1.6
8	7	10	0.9
		χ^2_{cal}	4.0

Since $\chi^2_{cal} < \chi^2_{tab}$, we fail to reject H_0
or Die is fair. There is no sufficient evidence at 5% significance level to suggest that die is not fair

5. Let X denote the number of flaws observed ... coil of galvanized steel. 75 coils are inspected and the foll data were observed for the values of X:

Values Observed	1	2	3	4	5	6	7	8
Frequency	1	11	8	13	11	12	10	9

(a) Does the assumption of Poisson distn. seem appropriate as a prob. model for this data? Use $\alpha = 0.01$.

b) Calculate P-value for this test.

H_0: ⇒ Poisson distn is appropriate for data

H_1: Data does not follow poisson distn

$\alpha = 0.01$.

$$\lambda = \frac{1*1 + 2*11 + 3*8 + 4*13 + 5*11 + 6*12 + 7*10 + 8*9}{75}$$

$$= 4.9067.$$

$$P(X=x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$P(X=1) = 0.036$ $P(4) = 0.1786$

$P(X=2) = 0.089$

$P(X=3) = 0.146$

$P(X=1)^{*75} = 2.7214$, $P(X=3)^{*75} = 10.9213$.

$P(X=2)^{*75} = 6.6770$ $P(X=4)^{*75} = 13.3977$

$P(X=5) = 13.1485$. $P(X=6) = 10.7533$.

$P(X=7) = 7.5381$ $P(X=8) = 4.6237$.

Values	O_i.	E_i.	$(O_i - E_i)^2 / E_i$.
1	1 } 12	2.7214 }	9.3987
2	11	6.6770	0.7200.
3	8	10.9213	0.7822
4	13	13.3977	0.0118
5	11	13.1485	0.3511

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6	12	10.7533		0.1445
7	10	4.5381	12.1618	3.8449
8.	9	4.6237		

$\chi^2_{cal} = 5.8545$.

$\chi^2_{tab} = \chi^2_{0.01, 6-1-1} = \chi^2_{0.01, 4} = 13.28$.

Since $\chi^2_{cal} < \chi^2_{tab}$ we fail to reject null hypothesis.
i-e The data follows Poisson. distn.

Pvalue for this test are

$0.1 < p\text{-value} < 0.50$